Predict First Qns:

1. Before you run this test, write down what you expect to happen if you delete your supplier variable (del supplier) after creating the products. Can each Product still display its supplier information? Run it and check. If your prediction did not match, note why.
2. Before step 3, write down whether you expect the Product to still work after it is removed from Inventory A, and why. Then check.
3. Predict first: If you add the same Product to two different Orders, how many OrderLine objects and how many Product objects will exist? Write your answer down, then check it (for example, compare objects using is or id()).
4. Before running your test, write down what each product will print for get_fulfilment_info(). Then compare with the actual output.

Predict First Ans:

1. Yes, each Product can still display its supplier information. This is because each Product already has a reference to the Supplier object. Deleting the supplier variable does not delete the actual Supplier object as the Products are still referring to it.
2. Yes, the Product will still work even after being removed from Inventory A. This is because removing a Product from an Inventory does not delete the Product itself. It can still exist on its own and be added to another Inventory, such as Inventory B.
3. There will be 2 OrderLine objects but only 1 Product object. This is because each Order creates its own OrderLine, but both OrderLines can refer to the same Product. Therefore, the Product is not duplicated.
4. The PhysicalProduct and DigitalProduct will print different fulfilment information. This is because each type of product has its own version of the get_fulfilment_info() method, so they will display information based on their product type.

Link to GitHub:

https://github.com/owentan654/INF1103-Labs.git

```
PS C:\Users\osmta\.vscode\INF1103-Labs\Lab 8> git remote -v
origin  https://github.com/owentan654/INF1103-Labs (fetch)
origin  https://github.com/owentan654/INF1103-Labs (push)
```

Docker Images:

```
PS C:\Users\osmta\.vscode\INF1103-Labs\Lab 8> docker images

IMAGE                                ID             DISK USAGE   CONTENT SIZE   EXTRA
inf1103-labs-smart-auditor:latest    0bf833b82d1f       198MB         48.5MB
inf1103-labs:latest                  19180fde448f       198MB         48.5MB
python:3.11-slim                     9534e5a8e315       200MB         50.7MB
```

Image runs:

```
PS C:\Users\osmta\.vscode\INF1103-Labs\Lab 8> docker run --rm -it -v "${PWD}:/app" -w /app inf1103-labs-smart-auditor:latest
No file found. Using default products.

===== Inventory Menu =====
1. Display all products
2. Add product
3. Update stock
4. Search Product
5. Remove Product
6. Create Order
7. Save Inventory
8. Exit
==========================
Choice:
```

```
Choice: 1

=========================================================
P001        Laptop              15          1200.00
            Weight: 2.0 kg
P002        Mouse               40          25.50
            Weight: 0.1 kg
P003        Antivirus           25          45.00
            Download size: 350 MB
=========================================================
```

```
Choice: 2
Type (1 = Physical, 2 = Digital): 1
Product ID: P004
Product name: Monitor
Price: 299.99
Stock: 10
Weight (kg): 5
Supplier ID: 1001
New supplier name: NVIDIA
Supplier email: NVIDIA@gmail.com
Product added.
```

```
Choice: 3
Product ID: P004
New stock quantity: 5
Stock updated.
```

```
Choice: 1

=====================================================
P001        Laptop              15          1200.00
            Weight: 2.0 kg
P002        Mouse               40          25.50
            Weight: 0.1 kg
P003        Antivirus           25          45.00
            Download size: 350 MB
P004        Monitor             5           299.99
            Weight: 5.0 kg
=====================================================
```

```
Choice: 4
Product ID: P004
P004        Monitor             5           299.99
            Weight: 5.0 kg
Fulfilment: Requires physical shipping.
Supplier 1001: NVIDIA < NVIDIA@gmail.com>
Shipping cost: $17.50
```

```
Choice: 5
Product ID: P004
Removed 'Monitor' from the inventory (the product object itself still exists).
```

```
=========================
Choice: 1

=====================================================
P001        Laptop              15          1200.00
            Weight: 2.0 kg
P002        Mouse               40          25.50
            Weight: 0.1 kg
P003        Antivirus           25          45.00
            Download size: 350 MB
=====================================================
```

```
Choice: 6
Order ID: 1
Product ID to add (blank to finish): P001
Quantity: 2
Product ID to add (blank to finish):
\Order 1
------------------------------------------------------
Laptop               x 2     @   1200.00 =    2400.00
------------------------------------------------------
Total:                                        2400.00
```

```
 Choice: 1

 ============================================================
 P001       Laptop              15          1200.00
            Weight: 2.0 kg
 P002       Mouse               40          25.50
            Weight: 0.1 kg
 P003       Antivirus           25          45.00
            Download size: 350 MB
 ============================================================
```

```
============================
Choice: 7
Inventory saved.
```

```
Choice: 8

==============================
Final Report:
==============================
Total units processed: 80 units.
Number of Failed/Rejected Entries: 0
Total tax to be paid: $8.00
==============================

Report saved to report.txt
```

```
============================================================
2. ASSOCIATION: Supplier and Product
============================================================
Supplier S100: Acme Supply < orders@acme.com>
Supplier S100: Acme Supply < orders@acme.com>
Same Supplier object shared by both products? True

After `del supplier`:
Supplier S100: Acme Supply < orders@acme.com>
Supplier S100: Acme Supply < orders@acme.com>
```

```
============================================================
3. AGGREGATION: Inventory and Product
============================================================
In A: ['Test Laptop']
Removed from A: Test Laptop
A now: []
Product still usable:
T1          Test Laptop          5          1000.00
            Weight: 2.0 kg
In B: ['Test Laptop']
Same object in B as the original? True
```

```
============================================================
4. COMPOSITION: Order and OrderLine
============================================================
\Order 01
------------------------------------------------------------
Test Laptop          x 2     @   1000.00 =     2000.00
Test Ebook           x 3     @     10.00 =       30.00
------------------------------------------------------------
Total:                                        2030.00

OrderLine objects equal?   False
Product objects equal?     True
Order 1 lines / Order 2 lines: 2 / 1

After removing the ebook line from Order 1:
\Order 01
------------------------------------------------------------
Test Laptop          x 2     @   1000.00 =     2000.00
------------------------------------------------------------
Total:                                        2000.00
Ebook product still exists: Test Ebook
```

```
============================================================
5. TYPE HINTS evidence
------------------------------------------------------------
Inventory.find_product: {'product_id': <class 'str'>, 'return': product.Product | None}
Inventory.add_product : {'product': <class 'product.Product'>, 'return': <class 'bool'>}
Product.__init__      : {'product_id': <class 'str'>, 'name': <class 'str'>, 'price': <class 'float'>, 'stock': <class 'int'>, 'supplier': <class 'supplier.Supplier'>, 're
turn': None}
Order.add_order_line  : {'product': <class 'product.Product'>, 'quantity': <class 'int'>, 'return': None}
PS C:\Users\usata\.vscode\INF1101-Labs>
```

```
============================================================
6/7. LSP + POLYMORPHISM through the Product type
============================================================
show_product_details() with each product type:
P1          Keyboard            25          45.00
            Weight: 0.8 kg
Requires physical shipping.
  update_stock(50) -> True | stock now 50
  update_stock(-1) -> False | stock now 50
P2          Antivirus           100         30.00
            Download size: 350.0 MB
Delivered digitally via download link.
  update_stock(50) -> True | stock now 50
  update_stock(-1) -> False | stock now 50
```

```python
section("2. ASSOCIATION: Supplier and Product")
supplier = Supplier("S100", "Acme Supply", "orders@acme.com")
laptop = PhysicalProduct("T1", "Test Laptop", 1000.0, 5, supplier, 2.0)
ebook = DigitalProduct("T2", "Test Ebook", 10.0, 5, supplier, 50.0)

laptop.get_supplier().display_information()
ebook.get_supplier().display_information()
print("Same Supplier object shared by both products?",
      laptop.get_supplier() is ebook.get_supplier())

del supplier
print("\nAfter `del supplier`:")
laptop.get_supplier().display_information()
ebook.get_supplier().display_information()
```

```python
section("3. AGGREGATION: Inventory and Product")
inv_a = Inventory()
inv_b = Inventory()
inv_a.add_product(laptop)                    # product existed BEFORE any inventory
print("In A:", [p.get_name() for p in inv_a.get_products()])

removed = inv_a.remove_product("T1")
print("Removed from A:", removed.get_name() if removed else None)
print("A now:", [p.get_name() for p in inv_a.get_products()])
print("Product still usable:")
laptop.display_information()

inv_b.add_product(laptop)                    # transfer to B
print("In B:", [p.get_name() for p in inv_b.get_products()])
print("Same object in B as the original?", inv_b.search_product("T1") is laptop)
```

```
section("4. COMPOSITION: Order and OrderLine")
order1 = Order("O1")
order1.add_order_line(laptop, 2)
order1.add_order_line(ebook, 3)
order1.display_information()

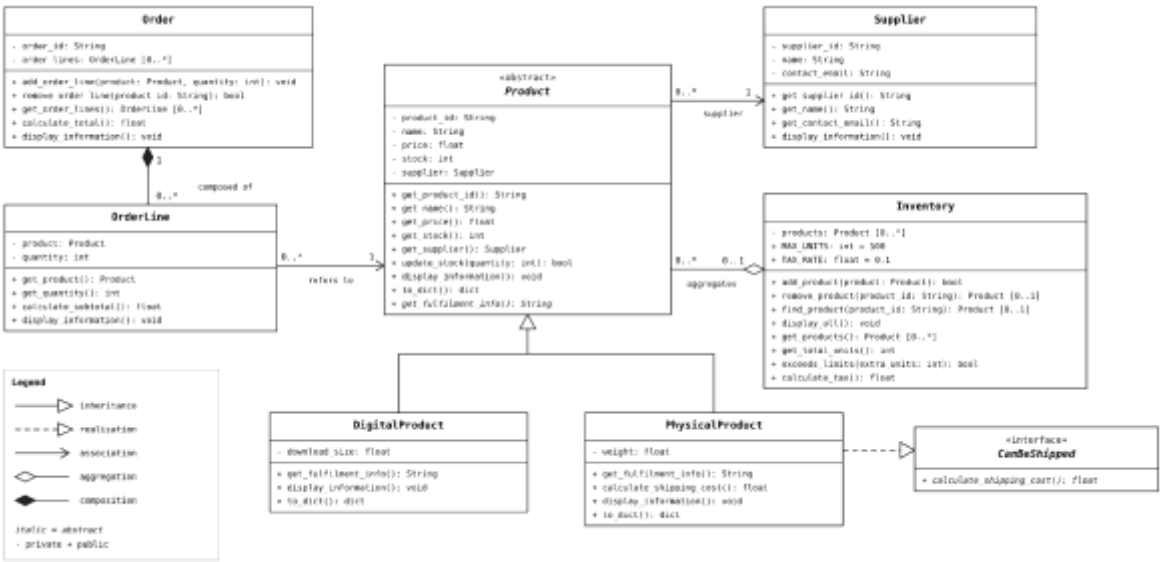
order2 = Order("O2")
order2.add_order_line(laptop, 1)              # same Product, different Order

line_in_1 = order1.get_orderlines()[0]
line_in_2 = order2.get_orderlines()[0]
print("\nOrderLine objects equal?  ", line_in_1 is line_in_2)
print("Product objects equal?    ", line_in_1.get_product() is line_in_2.get_product())
print("Order 1 lines / Order 2 lines:",
      len(order1.get_orderlines()), "/", len(order2.get_orderlines()))

order1.remove_order_lines("T2")
print("\nAfter removing the ebook line from Order 1:")
order1.display_information()
print("Ebook product still exists:", ebook.get_name())
```

```
section("6/7. LSP + POLYMORPHISM through the Product type")
inventory = Inventory()
supplier2 = Supplier("S200", "Beta Corp", "hi@beta.com")
inventory.add_product(PhysicalProduct("P1", "Keyboard", 45.0, 25, supplier2, 0.8))
inventory.add_product(DigitalProduct("P2", "Antivirus", 30.0, 100, supplier2, 350.0))

print("show_product_details() with each product type:")
for p in inventory.get_products():            # one loop, no isinstance()
    show_product_details(p)
    print("  update_stock(50) ->", p.update_stock(50), "| stock now", p.get_stock())
    print("  update_stock(-1) ->", p.update_stock(-1), "| stock now", p.get_stock())
```




```
PS C:\Users\osmta\.vscode\INF1103-Labs\Lab 8> docker history inf1103-labs-smart-auditor:latest
IMAGE          CREATED        CREATED BY                                      SIZE      COMMENT
0bf833b82d1f   2 weeks ago    CMD ["python" "OOP_auditor.py"]                 0B        buildkit.dockerfile.v0
```